AI ML — Zero to Master

24 topics, grouped as Beginner → Intermediate → Advanced → Master. Work through the topics in order; type every code example yourself and finish the practice tasks.

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Beginner Level

01. What are AI, ML and Deep Learning?

AI is machines performing tasks that need intelligence; machine learning learns patterns from data instead of explicit rules; deep learning uses multi-layer neural networks.

02. Python for Data Science Setup

Install Python, create a virtual environment and use Jupyter. Core libraries: NumPy, pandas, matplotlib, scikit-learn.

```
pip install numpy pandas matplotlib scikit-learn jupyter
```

03. NumPy

Fast n-dimensional arrays and vectorised maths.

```
import numpy as np
a = np.array([1, 2, 3])
print(a.mean(), a * 2)
```

04. pandas: Loading and Cleaning Data

DataFrames for loading CSVs, handling missing values, filtering and grouping.

```
import pandas as pd
df = pd.read_csv('data.csv')
df = df.dropna()
df.groupby('city')['sales'].mean()
```

05. Visualisation

Plot with matplotlib/seaborn to understand distributions and relationships.

```
df['price'].hist()
import seaborn as sns; sns.scatterplot(data=df, x='size', y='price')
```

06. Statistics and Math Basics

Mean, median, variance, probability, correlation, linear algebra, and derivatives for optimisation.

Intermediate Level

07. Machine Learning Workflow

Problem -> data -> clean -> features -> split train/test -> train -> evaluate -> deploy -> monitor.

```
from sklearn.model_selection import train_test_split
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, random_state=42)
```

08. Linear and Logistic Regression

Predict numbers (regression) or classes (classification).

```
from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X_tr, y_tr)
model.score(X_te, y_te)
```

09. Decision Trees and Random Forests

Tree-based models handle non-linear data; ensembles reduce overfitting.

```
from sklearn.ensemble import RandomForestClassifier
clf = RandomForestClassifier().fit(X_tr, y_tr)
```

10. Evaluation Metrics

Accuracy, precision, recall, F1, ROC-AUC for classification; MAE, RMSE, R2 for regression; confusion matrix.

```
from sklearn.metrics import classification_report
print(classification_report(y_te, clf.predict(X_te)))
```

11. Overfitting, Cross-validation and Tuning

Detect overfitting; use cross-validation and `GridSearchCV` for hyper-parameters.

```
from sklearn.model_selection import cross_val_score
cross_val_score(clf, X, y, cv=5)
```

12. Feature Engineering and Pipelines

Scaling, encoding categories, handling imbalance, and `Pipeline` to avoid data leakage.

```
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
make_pipeline(StandardScaler(), LogisticRegression())
```

13. Unsupervised Learning

Clustering (K-means), dimensionality reduction (PCA), anomaly detection.

```
from sklearn.cluster import KMeans
KMeans(n_clusters=3).fit(X)
```

Advanced Level

14. Neural Networks Fundamentals

Neurons, layers, activation functions, loss, backpropagation and gradient descent.

15. Deep Learning with PyTorch or TensorFlow/Keras

Build and train networks with tensors, autograd and optimisers.

```
import torch.nn as nn
model = nn.Sequential(nn.Linear(10, 32), nn.ReLU(), nn.Linear(32, 1))
```

16. Computer Vision: CNNs

Convolutional networks for image classification and detection; transfer learning.

17. NLP and Transformers

Tokenisation, embeddings, attention and transformer models; Hugging Face.

```
from transformers import pipeline
pipeline('sentiment-analysis')('I love this course')
```

18. Large Language Models and Prompting

How LLMs work, prompt engineering, context windows, temperature, evaluation and safety.

19. RAG and Vector Databases

Retrieval-Augmented Generation: embed documents, search a vector DB (pgvector, FAISS, Pinecone), and feed context to an LLM.

20. Working with LLM APIs and Agents

Call model APIs, structured outputs, tool use / function calling and agents.

```
from anthropic import Anthropic
client = Anthropic()
client.messages.create(model='claude-sonnet-5-5', max_tokens=200, messages=[{'role': 'user',
'content': 'Hello'}])
```

Master Level

21. MLOps: Deployment and Monitoring

Package models, serve with FastAPI/Docker, track experiments (MLflow), CI/CD, drift monitoring.

22. Responsible AI, Bias and Ethics

Fairness, privacy, explainability (SHAP), safety and regulation.

23. Fine-tuning and Model Optimisation

LoRA/PEFT fine-tuning, quantisation, distillation and evaluation harnesses.

24. Real Projects

House price prediction, spam classifier, image classifier, chatbot with RAG, recommendation system.

Interview Question Bank

Q1. AI vs ML vs DL?
AI is the broad goal; ML learns from data; DL is ML using deep neural networks.

Q2. Supervised vs unsupervised learning?
Supervised uses labelled data; unsupervised finds structure in unlabelled data.

Q3. What is overfitting?
The model memorises training data and performs poorly on new data; reduce with regularisation, more data, simpler models.

Q4. Bias-variance trade-off?
High bias underfits; high variance overfits; aim for balance.

Q5. Precision vs recall?
Precision = correct positives / predicted positives; recall = correct positives / actual positives.

Q6. What is gradient descent?
An optimisation algorithm that adjusts parameters in the direction that reduces the loss.

Q7. What is a train/validation/test split?
Train to learn, validation to tune, test for a final unbiased score.

Q8. What is a transformer?
A neural architecture using self-attention, the foundation of modern LLMs.

Q9. What is RAG?
Retrieval-Augmented Generation: retrieving relevant documents to ground an LLM's answer.

Q10. What is data leakage?
Information from outside the training set leaking into training, inflating results.

Capstone Project Ideas

1. House price predictor
2. Spam / sentiment classifier
3. Image classifier with transfer learning
4. Chatbot with RAG over PDFs
5. Movie recommender